

23-S1-Q2

Q : Data preparation

ca) data types + example

cb) outliers

cc) missing value

cd) normalization

Solution (a)

① Qualitative, Nominal

Examples: Gender (Male, Female)

Name (Zhang San, Li Si)

Country (China, USA)

② Qualitative, Ordinal

Examples: Customer-satisfaction scale (Happy > unhappy)

Academic grade ($A^+ > A > B > C$)

③ Quantitative, Interval

Examples: Celsius temperature

Calendar years

④ Quantitative, Ratio

Examples: Height (cm)

weight (kg)

salary (USD)

(b) ① Remove outliers : If the number of outliers is not many , a simple approach may be to remove them

② Imputation : Impute the value with mean or median or mode.

③ Capping : For values that lie outside the 1.5 times of IQR limits, we cap them by replacing those observations with the value of α th percentile or the value of 95th percentile

④ If there is a significant number of outliers, they should be treated separately in the statistical model

⑤ If the outlier are natural, we shouldn't amend it

- (4) ① remove the samples having missing value
② Imputing missing value

(1) For quantitative attribute, missing values can be imputed with the mean, median or mode of the remaining values under the same attribute

(2) For quantitative attribute : mode.

(3) Another strategy is to identify the similar types of observations whose values are known and use the mean/median/mode of those known values.

(d) Feature Scaling

① Normalization (Min-Max scaling)

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \rightarrow [0, 1]$$

② Standardization

$$x' = \frac{x - \mu}{\sigma} \rightarrow \text{mean } 0, \text{SD } 1$$